Loan Eligibility Policy Recommendation

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Title

Loan Eligibility Policy Recommendation

Objective

Use a simple EMI + FOIR + credit-score rule as an educational decision-support screen.

Current baseline

Eligible when FOIR <= 50% AND credit score >= 700.

Evidence

500 synthetic applications; 43 actual defaults and 457 non-defaults.

Baseline result

203 eligible, 297 not eligible; 9 defaults accepted; 263 good applicants rejected; accuracy 45.6%.

Tested alternatives

25 combinations: FOIR thresholds 40/45/50/55/60% crossed with credit-score thresholds 650/675/700/725/750.

Operating recommendation

Retain the 50% / 700 baseline as the primary operating rule in this teaching dataset; use 55% / 700 only as an explicitly monitored higher-approval alternative.

Why

The 55% / 700 alternative increases eligible cases from 203 to 224 and reduces good applicants rejected from 263 to 243, but also increases defaults accepted from 9 to 10. This is a clear approval-vs-risk trade-off rather than a universally better rule.

Risk-conservative alternative

50% / 725 accepts 6 defaults but rejects 318 good applicants; it is materially stricter and sacrifices approvals.

Responsible AI note

This analysis uses synthetic teaching data. No real customer information was used. AI-assisted content/calculations were independently checked. The calculator is an educational prototype and should not be treated as a production credit underwriting system. Real credit decisions require human review, privacy, fairness, explainability, regulatory controls and monitoring.